Automatic GFL/GFM Mode Switching

Mr. Phichitchai Jueajan (66010569)
Mr. Pakkapol Phdungdan (66010612)

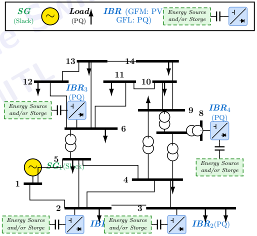
Advisor: Dr. Tossaporn Surinkaew

Department of Electrical Engineering, Faculty of Engineering
King Mongkut's Institute of Technology Ladkrabang (KMITL)

Table of Contents

Introduction

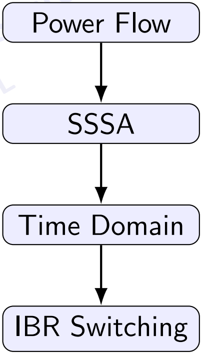
- Increasing IBR penetration reduces reliance on synchronous generation.
- GFL converters require an external voltage-angle reference.
- GFM converters can establish voltage and frequency in weak-grid or islanded operation.
- This project studies automatic GFL/GFM switching in an IEEE 14-bus system with one SG and four dual-mode IBRs.



Research focus: maintain a valid grid reference after SG loss and return support after SG recovery.

Project Objectives

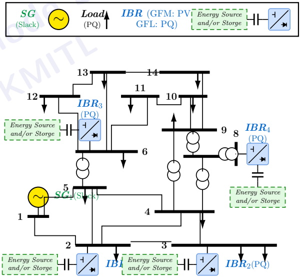
1. Develop an in-house analysis chain for Power Flow, SSSA, and Time-Domain Simulation.
2. Model four IBRs that can operate in either GFL or GFM mode.
3. Select and switch GFM support according to the system condition.
4. Verify switching, island operation, and synchronous-generator reconnection under one disturbance chronology.



Study System: IEEE 14-Bus with 1 SG + 4 IBRs

- SG1 at bus 1 provides the normal grid reference.
- Dual-mode IBRs are connected at buses 2, 3, 6, and 8.
- Each IBR supports **GFL**, **GFM**, and tripped states.
- The network equations are shared by PF, SSSA, and TS.

$$\boxed{\boldsymbol{g}(\boldsymbol{x},\boldsymbol{y}) = \boldsymbol{Y}_{net}\boldsymbol{V} - \boldsymbol{I}_{inj}(\boldsymbol{x},\boldsymbol{V}) = \boldsymbol{0}}$$



Power Flow Formulation

For bus i,

$$P_i = V_i \sum_{k=1}^{n} V_k \left(G_{ik}\cos\theta_{ik} + B_{ik}\sin\theta_{ik}\right)$$

$$Q_i = V_i \sum_{k=1}^{n} V_k \left(G_{ik}\sin\theta_{ik} - B_{ik}\cos\theta_{ik}\right)$$

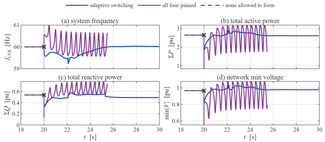
Newton–Raphson iteration:

$$\boxed{\boldsymbol{J}_{PF}(\boldsymbol{z}^{(r)})\Delta\boldsymbol{z}^{(r)} = -\boldsymbol{F}_{PF}(\boldsymbol{z}^{(r)})}$$

$$\boxed{\boldsymbol{z}^{(r+1)} = \boldsymbol{z}^{(r)} + \Delta\boldsymbol{z}^{(r)}}$$

Bus constraints

Bus	Specified	Solved
REF	$\lvert V\rvert, \theta$	P, Q
PV	$P, \lvert V\rvert$	Q, θ
PQ	P, Q	$\lvert V\rvert, \theta$



Small-Signal Stability Analysis (SSSA)

Nonlinear DAE:

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{y}), \qquad \boldsymbol{0} = \boldsymbol{g}(\boldsymbol{x}, \boldsymbol{y})$$

Linearization around $(\boldsymbol{x}_0, \boldsymbol{y}_0)$:

$$\begin{bmatrix} \Delta\dot{\boldsymbol{x}} \\ \boldsymbol{0} \end{bmatrix} = \begin{bmatrix} \boldsymbol{f}_x & \boldsymbol{f}_y \\ \boldsymbol{g}_x & \boldsymbol{g}_y \end{bmatrix} \begin{bmatrix} \Delta\boldsymbol{x} \\ \Delta\boldsymbol{y} \end{bmatrix}$$

Eliminating algebraic variables:

$$\boxed{\boldsymbol{A} = \boldsymbol{f}_x - \boldsymbol{f}_y(\boldsymbol{g}_y \backslash \boldsymbol{g}_x)}$$

$$\boxed{\boldsymbol{A}\boldsymbol{v} = \lambda\boldsymbol{v}}$$

$$f = \frac{|\operatorname{Im}\lambda|}{2\pi}, \qquad \zeta = -\frac{\operatorname{Re}\lambda}{|\lambda|}$$

Stability criterion

$\operatorname{Re}(\lambda) < 0$ for all physical modes; the selector also checks the declared damping-ratio floor.

Time-Domain Simulation (TS)

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{y}), \qquad \boldsymbol{0} = \boldsymbol{g}(\boldsymbol{x}, \boldsymbol{y})$$

Implicit trapezoidal step

$$\boldsymbol{R}_x = \boldsymbol{x}_{n+1} - \boldsymbol{x}_n - \frac{h}{2}(\boldsymbol{f}_n + \boldsymbol{f}_{n+1}) = \boldsymbol{0}$$

$$\boldsymbol{R}_y = \boldsymbol{g}(\boldsymbol{x}_{n+1}, \boldsymbol{y}_{n+1}) = \boldsymbol{0},$$

$$\boxed{\boldsymbol{J}_R \Delta \boldsymbol{z}} = \boxed{-\boldsymbol{R}}, \qquad \boldsymbol{z} = [\boldsymbol{x}; \boldsymbol{y}].$$

Event handling

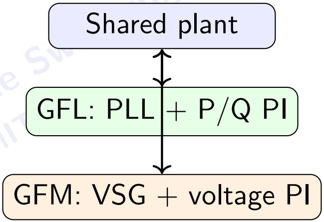
- Exact landing on event times
- Preserve the left-limit state
- Atomic mode/topology transaction
- Right-limit KCL must converge
- Roll back a rejected transaction

Dual-Mode IBR State Representation

Each converter uses a fixed 17-coordinate superset:

$$\boxed{\boldsymbol{x}_{IBR} = \left[\boldsymbol{x}_p^T \quad \boldsymbol{x}_{GFL}^T \quad \boldsymbol{x}_{GFM}^T \quad I_{dc}\right]^T}$$

$$\begin{aligned}
\boldsymbol{x}_p &= [i_d \; i_q \; V_{dc}]^T, \\
\boldsymbol{x}_{GFL} &= [\delta_{PLL} \; \xi_{PLL} \; \xi_P \; \xi_Q \; \xi_{Id} \; \xi_{Iq}]^T, \\
\boldsymbol{x}_{GFM} &= [\delta_{VSG} \; \omega_{VSG} \; E \; \xi_{Vd} \; \xi_{Vq} \; \xi_{Id}^{GFM} \; \xi_{Iq}^{GFM}]^T \\
\mathcal{A}_{GFL} &= \{1{:}9, 17\}, \qquad \mathcal{A}_{GFM} = \{1{:}3, 10{:}17\}.
\end{aligned}$$



Mode rule

Only the selected controller branch evolves; inactive controller states are held.

GFL and GFM State Equations

Grid-Following (GFL)

$$\begin{aligned}
\dot{\delta}_{PLL} &= k_{p,PLL} v_q + k_{i,PLL} \xi_{PLL}, \\
\dot{\xi}_{PLL} &= v_q, \\
e_P &= \kappa P^\star - P, \\
e_Q &= \kappa Q^\star - Q, \\
i_d^\star &= k_{pP} e_P + k_{iP} \xi_P, \\
i_q^\star &= -(k_{pQ} e_Q + k_{iQ} \xi_Q).
\end{aligned}$$

PLL-synchronized current control

Grid-Forming (GFM)

$$\begin{aligned}
M\dot{\omega} &= \kappa P^\star - P - D_v(\omega - 1), \\
\dot{\delta}_{VSG} &= \omega_b(\omega - 1), \\
\tau_E \dot{E} &= k_Q(\kappa Q^\star - Q) - k_E(E - E^\star), \\
e_{vd} &= E - v_d, \qquad e_{vq} = -v_q, \\
i_d^\star &= k_{pV} e_{vd} + k_{iV} \xi_{Vd}, \\
i_q^\star &= k_{pV} e_{vq} + k_{iV} \xi_{Vq}.
\end{aligned}$$

VSG establishes angle and frequency

Automatic GFL/GFM Switching Law

Severity index

$$J_V = \frac{\big||V_i| - V_i^\star\big|}{0.10\,\mathrm{p.u.}},$$
$$J_f = \frac{|f_{COI} - 60|}{0.50\,\mathrm{Hz}},$$
$$S = \mathrm{sat}_{[0,1]}(0.5J_V + 0.5J_f)\,.$$

Mode command

$$q^+ = \begin{cases} \mathrm{GFM}, & S \geq 0.65 \quad (0.10\,\mathrm{s}), \\ \mathrm{GFL}, & S < 0.35 \quad (1.00\,\mathrm{s}), \\ q, & \text{otherwise.} \end{cases}$$

Switch transaction

$$\boxed{\left\| I^+ - I^- \right\|_\infty \leq 10^{-10}\,\mathrm{p.u.}}$$

- Equilibrium + SSSA certified candidate
- Target states initialized at the present terminal point
- Right-limit KCL converges before commit

Result 1: GFM Candidate Screening

GFM count	Selected buses	Outcome
1	2, 3, 6, or 8	admissible
2	2+6	admissible
2	**3+6**	**best margin**
2	four other pairs	rejected
3	all triples	rejected
4	2+3+6+8	admissible

Admissibility is non-monotonic in GFM count.

Selector sequence

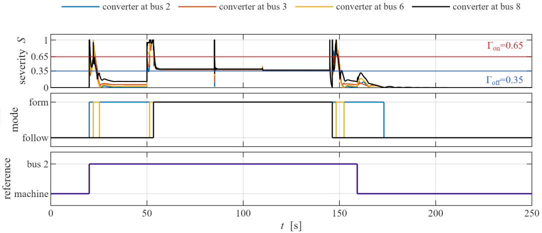
Topology → SCR → equilibrium → full-KCL SSSA → authenticated candidate

The GFM set is computed for the network condition; it is not chosen by a "more GFM is better" rule.

Result 2: Supervisor and Mode Switching

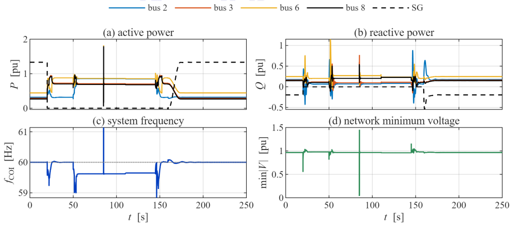
- SG trip: 20 s
- Load step: 50 s
- Bus-9 fault: 85–85.15 s
- Line 6–13 trip: 110 s
- Restore / SG reclose request: 145 s

Policy: add GFM support during severe conditions and release it after recovery.



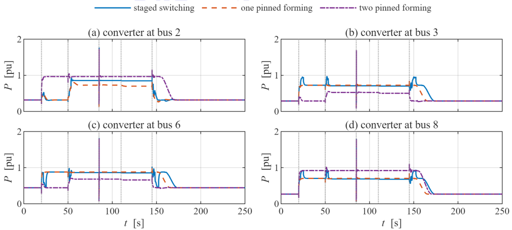
Result 3: Electrical Response

- IBRs support the island after the SG trip.
- Frequency and voltage respond to each scheduled disturbance.
- The SG is reconnected only after the synchronism conditions are satisfied.
- The accepted run reaches the full 250 s horizon.



Result 4: Converter Response under Different Policies

- Staged switching: solid lines
- One pinned GFM: dashed lines
- Two pinned GFMs: dash-dot lines

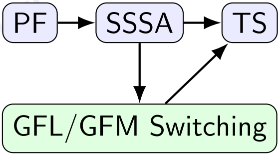


Main observation

The power-sharing trajectory depends strongly on which converters are forming the grid and on when each mode transition occurs.

Conclusion

- PF establishes the nonlinear operating point.
- SSSA screens candidate mode configurations around their equilibria.
- TS verifies the nonlinear trajectory through disturbances and switching.
- At least one GFM is required after the synchronous reference is lost.
- GFM admissibility is not monotonic with the number of forming units.
- Staged switching completes the 250 s chronology and returns IBR support after SG recovery.



Thank You